

PROFILE

I work at the intersection of machine learning, scientific inquiry, and engineering design. My current interests include AI for Science, foundation models for technical workflows, and AI-assisted analog circuit modeling and optimization.

HIGHER EDUCATION

University of Minnesota May 2026 — Present

B.S. studies in Computer Science and Electrical Engineering

Delft University of Technology × Eindhoven University of Technology September 2024 — August 2026

Concurrent B.Sc. studies across Computer Science & Engineering, Electrical Engineering, and Applied Physics

Transferred to the University of Minnesota.

RESEARCH EXPERIENCE

NLP Research Intern · University of Minnesota — Minnesota NLP Group July 2026 — Present

Undergraduate research internship in natural language processing. Public project details are linked only when supporting evidence is available.

Research Assistant · The Chinese University of Hong Kong March 2026 — Present

Research activity in LLM for Recommendation (LLM4Rec) and AI for Science.

Research Assistant · Zhangjiang Laboratory July 2025 — September 2025

Studied semiconductor IC design workflows and EDA toolchains from schematic and pre-layout simulation through verification and MPW-oriented tape-out preparation.

- Documented public PDK concepts and how DRC constraints influence layout strategy and manufacturability.
- Compiled notes on DRC/LVS roles, common failure modes, and sign-off debugging approaches.
- Prepared a milestone and risk overview from schematic sign-off to foundry submission; no tape-out claim is made.

Student Researcher · Independent research project September 2023 — February 2024

Investigated PM2.5 exposure and respiratory-disease incidence under the joint supervision of Prof. Soumya Kar (Carnegie Mellon University) and Associate Prof. Qiang Gao (Southeast University).

- Built a structured environmental and public-health dataset and used Python for exploratory analysis, statistical inference, and visualization.

TEACHING

Teaching Assistant ·
Eindhoven University
of Technology

September 2025 — November 2025

Supported undergraduate teaching in calculus and logic/set theory through tutorials, problem solving, and structured feedback.

- Guided work on limits, derivatives, integrals, formal reasoning, and set operations.

ENTREPRENEURSHIP

Co-Founder · Veylora Labs

May 2026 — Present

Developing executive-intelligence systems that help organizational leaders identify patterns, risks, and opportunities across fragmented operational information.

RESEARCH INTERESTS

AI for Science: Learning systems for scientific and engineering problems where physical structure, constraints, and reproducible evidence matter.

Foundation models: Foundation models as reasoning tools for recommendation, structured knowledge, and complex technical workflows.

AI-driven analog EDA: Machine-learning methods for analog circuit modeling, optimization, and design automation under physical constraints.

SELECTED PUBLIC EVIDENCE

- 01 topo-flow-limits
appleweiping/topo-flow-limits
Research code examining when latent higher-order network structure can be identified from edge-flow observations.
- 02 inr-aliasing-limits
appleweiping/inr-aliasing-limits
Research code examining identifiability and anti-aliasing sample design for implicit neural representations with fixed Fourier features.
- 03 Lumen-Rec
appleweiping/Lumen-Rec
Research code for comparing pointwise LLM relevance estimates and uncertainty-aware reranking under a shared candidate protocol.
- 04 TRUCE-Rec
appleweiping/TRUCE-Rec
Research code for uncertainty-aware generative recommendation, organized around shared data, baselines, evaluation, and reproducibility utilities.
- 05 venus-basestation
appleweiping/venus-basestation
Coursework base-station software with a live visualization dashboard for a multi-robot planetary-exploration project.
- 06 art-history-museum
appleweiping/art-history-museum
A browser-based interactive 3D art museum with a navigable history timeline and first-person galleries.

METHODS & SKILLS

Statistical inference

Data analysis

Data visualization

Machine learning

Scientific computing

Research software engineering

EARLIER EDUCATION

The Affiliated High School to
Hangzhou Normal University

September 2021 — August 2024

High School Diploma — Chinese, Mathematics, English, Physics, Chemistry, and Technology

Shanghai Foreign Language
School Affiliated to SISU

July 2023 — August 2024

Science and Engineering